ZEROS OF POLYANALYTIC POLYNOMIALS: LINEAR BOUNDS AND EXPLICIT CONSTRUCTIONS

ANTONIO LERARIO, ERIK LUNDBERG, AND TENG ZHANG

Abstract. We prove that a polynomial of degree n in z and degree $m < n$ in $\bar{z}$, with finitely many zeros, has at most $\min\{n^2 + m^2, (4m+1)n - 5m\}$ distinct zeros. The coefficient $4m+1$ is optimal for every fixed m, and the maximal count differs from this upper bound by at most $4m(m-1)$. For harmonic polynomials of analytic degree n and conjugate degree m, we construct $(m+2)n - m - \gcd(m,n)$ nondegenerate zeros. These examples attain the known sharp bounds when $m = 1$ and $m = n-1$. We also determine the exact maximum in the family $\varepsilon z^n + q(z) + \overline{q(z)}$, and give sparse polyanalytic polynomials attaining the quadratic bound.

1. Introduction

The fundamental theorem of algebra counts the zeros of a polynomial in one complex variable by its degree. For a polynomial in z and $\bar{z}$, the zero set can instead be empty, finite, or contain a curve. Even when it is finite, the appropriate bound depends on which degrees are prescribed. We study the effect of fixing the degree in the conjugate variable.

For a map $F : \mathbb{C} \to \mathbb{C}$, write $N(F) = \#F^{-1}(0)$. For $f \in \mathbb{C}[z, w]$, put $F_f(z) = f(z, \bar{z})$ and abbreviate $N(F_f)$ to $N(f)$. All zero counts count distinct points. A zero of a smooth map is *nondegenerate* if its real differential is invertible. We say that f has bidegree (n, m) if $\deg_z f = n$ and $\deg_w f = m$, in that order. Total degree will always be specified. Let $M_{m,n}$ be the largest finite value of $N(f)$ among polynomials of bidegree (n, m), where $n > m \geq 1$. The subscript m records the conjugate degree, which we regard as the fixed parameter.

Khavinson, Lundberg, and Perry [KLP24, Problem 1.9 and Proposition 1.10] formulated the general valence problem for polyanalytic polynomials and proved the bound $n^2 + m^2$ when f and $f^*(z, w) = \overline{f(\bar{w}, \bar{z})}$ are coprime. Eremenko [Ere26, p. 1] asked whether the growth is linear in n for fixed m. The finite-zero-set formulation is as follows.

Problem 1. *For each $m \geq 1$, do there exist constants A_m, B_m such that $N(f) \leq A_m n + B_m$ for every polynomial of bidegree (n, m), $n > m$, with finite zero set?*

Finiteness is essential: $z^{n-m}(zw - 1)^m$ vanishes on the unit circle. Also, n in Problem 1 is not the total degree. For example, $(z^2 - a^2)\bar{z} - z$, with $0 < a < 1$, has bidegree $(2, 1)$ and the five distinct zeros 0, $\pm\sqrt{1 + a^2}$, $\pm i\sqrt{1 - a^2}$. Thus n^2 is not an upper bound in the rectangular coefficient space.

Our first result answers Problem 1 and determines its optimal leading constant.

Theorem 2. *For every $n > m \geq 1$,*

$$M_{m,n} \leq U_{m,n} := \min\{n^2 + m^2, (4m+1)n - 5m\}. \tag{1}$$

For each fixed $m \geq 1$, one has $\lim_{n\to\infty} M_{m,n}/n = 4m + 1$.

The estimate includes degenerate zeros and polynomials with common factors with f^*. Its proof first replaces such zeros by nondegenerate zeros without increasing either degree. The complex curve $f = 0$ in $\mathbb{CP}^1 \times \mathbb{CP}^1$ then gives an immersed surface in $\mathbb{CP}^2$. Resolving its double points and applying the genus theorem of Kronheimer and Mrowka [KM94] yields the linear term in (1). Eremenko's maintained problem list [EreList] records a solution of the growth question by this reduction; the proof and the optimality assertion are given here in full.

The lower bounds can be made uniform in both degrees. We use $x_+ = \max\{x, 0\}$ for a real number x.

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Theorem 3. *For every $n > m \geq 1$,*

$$U_{m,n} - 4m(m-1) \leq M_{m,n}. \tag{2}$$

One has $M_{m,n} = n^2 + m^2$ when $m < n \leq 2m+1$. For $n \geq 2m+1$, there are polynomials of exact bidegree (n, m) with exactly

$$(4m+1)(n-m) + (3m-n)_+(3m+1-n)_+ \tag{3}$$

nondegenerate zeros. The constructions may be chosen with coefficients in $\mathbb{Q}(i)$.

In particular, (1) and (2) give $M_{1,n} = 5n-5$. This endpoint is consistent with the theory of rational harmonic functions developed by Khavinson and Neumann [KN06], together with the sharp examples of Rhie [Rhi03]. For general m, Theorem 3 leaves an additive error depending only on m; it does not determine every finite-degree maximum.

1.1. **Harmonic polynomials.** Let $H_{m,n}$ denote the largest zero count of $h(z) = p(z) + \overline{q(z)}$, where $\deg p = n > \deg q = m$. Subtracting a constant preserves these degrees, so this is also the maximal valence in this class. The harmonic valence problem was raised by Sheil-Small [SS92, p. 19] in 1992; see also Khavinson and Świątek [KS03]. Wilmshurst [Wil94, Wil98] proved $H_{m,n} \leq n^2$ and constructed examples attaining this bound for $m = n-1$. An independent extremal construction was given by Bshouty, Hengartner, and Suez [BHS95]; Peretz and Schmid [PS97] gave an algebraic treatment of the quadratic bound. For $m = 1$, Khavinson and Świątek [KS03] proved the upper bound $3n-2$, and Geyer [Gey08] proved sharpness through holomorphic dynamics.

Wilmshurst [Wil98, Remark 2] proposed the more precise expression $3n-2+m(m-1)$. Lee, Lerario, and Lundberg [LLL15] disproved it and proposed linear growth for each fixed m. Further constructions were obtained by Hauenstein, Lerario, Lundberg, and Mehta [HLLM15], Lee and Saez [LS17], and Khavinson, Lee, and Saez [KLS18]. Lundberg [Lun23] proved $H_{m,n} \geq \lceil n\sqrt{m} \rceil$ by a probabilistic argument and asked for constructions with order mn zeros. Gao and Liu [GL26] recently obtained further examples at $m = n-1$ using alternating analytic and anti-analytic terms. Theorem 2 proves the fixed-m linear upper bound in the larger polyanalytic class. The following examples give a lower bound of order mn within the harmonic class itself.

Let T_j be the Chebyshev polynomial defined by $T_j(\cos\theta) = \cos(j\theta)$.

Theorem 4. *Let $1 \leq m < n$ and $a > 2$. For all sufficiently small $\varepsilon > 0$, the harmonic polynomial*

$$h_\varepsilon(z) = \varepsilon T_n(z) - 2\operatorname{Im} T_m(z-a) \tag{4}$$

has exactly $(m+2)n - m - \gcd(m,n)$ nondegenerate zeros.

Here $p(z) = \varepsilon T_n(z) + iT_m(z-a)$ and $q(z) = iT_m(z-a)$ give $h_\varepsilon = p + \bar{q}$. Both degrees are exact, and rational choices of a and ε give coefficients in $\mathbb{Q}(i)$. Theorem 4 implies

$$(m+2)n - m - \gcd(m,n) \leq H_{m,n} \leq \min\{n^2, (4m+1)n - 5m\}. \tag{5}$$

The lower bound in (5) is at least mn, and it recovers the sharp counts $3n-2$ and n^2 at $m = 1$ and $m = n-1$, respectively.

The case $m = 1$ has a particularly simple explicit form.

Corollary 5. *For every $n \geq 2$, the polynomial $p_n(z) = z - iT_n(z)/n^2$ satisfies $\#\{z : p_n(z) = \bar{z}\} = 3n-2$, and every zero of $p_n(z) - \bar{z}$ is nondegenerate.*

We prove Corollary 5 directly, without any divisibility argument. The n real zeros of T_n are joined by two nonreal zeros for each of its $n-1$ conjugate pairs of branches of $\operatorname{Im} T_n = 0$. We also determine every transition in the zero count of $\varepsilon T_n(z) - 2\operatorname{Im} z$, including the singular parameters.

Lundberg's question can be answered in the more restrictive family used in his probabilistic construction. We state the question in the form relevant to the present result.

Problem 6. *For $1 \leq m < n$, choose $\deg q = m$ and $\varepsilon \neq 0$ deterministically so that $\varepsilon z^n + q(z) + \overline{q(z)}$ has at least mn distinct zeros.*

Problem 6 is posed by Lundberg [Lun23, Section 4.2]. Define $S_{m,n}$ to be the maximal zero count in its displayed family and, with $g = \gcd(m,n)$, put $\delta_{m,n} = g$ if m/g and n/g are both odd, and $\delta_{m,n} = 0$ otherwise.

Theorem 7. *For every $1 \leq m < n$,*

$$S_{m,n} = n(m+1) + \delta_{m,n}. \tag{6}$$

The maximum is attained with $\varepsilon \in \mathbb{Q}_{>0}$ and $q \in \mathbb{Q}(i)[z]$, with all zeros nondegenerate.

The exact value in (6) concerns this particular family. Neither it nor Theorem 4 asserts an exact formula for $H_{m,n}$ at all degrees. In particular, the stronger conjecture of Lee, Lerario, and Lundberg [LLL15] that the number of orientation-reversing zeros is at most $m(n-1)$ is not resolved here.

1.2. **Sparse extremizers and organization.** Sète and Zur [SZ24, Remark 4.4] asked how many nonzero coefficients are necessary for a polynomial of total degree d to have d^2 zeros, and proposed the following conjecture.

Conjecture 8. *A polyanalytic polynomial of total degree d with d^2 distinct zeros has at least d nonzero coefficients.*

The radial constructions used for Theorem 3 also give counterexamples to Conjecture 8.

Theorem 9. *For every $d \geq 2$, there is a polynomial of total degree d with coefficients in $\mathbb{Q}$, exactly $\lceil d/2 \rceil + 1$ nonzero monomials, and d^2 nondegenerate zeros. The least possible number of monomials in total degree four is three, attained by*

$$z^4 + z\bar{z}^2 - \tfrac{1}{8}\bar{z}. \tag{7}$$

Section 2 gives the harmonic Chebyshev construction, beginning with the explicit $m = 1$ example. Section 3 proves the perturbation and intersection lemmas, and Section 4 proves the linear upper bound. Section 5 supplies the radial constructions and proves Theorems 3 and 9. Section 6 proves the exact restricted harmonic maximum. Section 7 records the remaining questions. All proofs in the paper are independent of numerical certification. The supplementary repository [LLZ26] contains reproducible checks, figures, a proof audit, and the original research archive.

2. Chebyshev polynomials and harmonic zeros

2.1. **The real-level curves.** The identity $T_n(\cos\zeta) = \cos(n\zeta)$ holds for complex ζ, by the polynomial recurrence for T_n. Every point of the upper half-plane has a unique representation

$$z = \cos(\theta - it) = \cos\theta\cosh t + i\sin\theta\sinh t, \qquad 0 < \theta < \pi, \quad t > 0. \tag{8}$$

Indeed, of the two solutions of $\xi + \xi^{-1} = 2z$, exactly one has modulus greater than one. Its imaginary part is positive, and writing it as $e^{t+i\theta}$ gives (8). The differential is nonsingular: its determinant is $-(\sin^2\theta + \sinh^2 t) < 0$. Since $\operatorname{Im} T_n(z) = \sin(n\theta)\sinh(nt)$, the nonreal upper branches of the real-level set are

$$z_j(t) = \cos\theta_j\cosh t + i\sin\theta_j\sinh t, \qquad \theta_j = \frac{j\pi}{n}, \quad 1 \leq j < n, \quad t > 0. \tag{9}$$

The remaining branches are their complex conjugates and the real axis. On the j-th branch, $T_n(z_j(t)) = (-1)^j\cosh(nt)$.

2.2. **A complete count when the conjugate degree is one.** For $m = 1$, the real translation in (4) disappears: $\operatorname{Im} T_1(z-a) = \operatorname{Im} z$. Put $\phi_n(t) = \sinh t/\cosh(nt)$ for $t > 0$. Its logarithmic derivative is $\coth t - n\tanh(nt)$, whose derivative is $-\operatorname{csch}^2 t - n^2\operatorname{sech}^2(nt) < 0$. The logarithmic derivative tends to $+\infty$ at zero and to $1 - n < 0$ at infinity. Thus there is a unique $t_n > 0$ satisfying

$$\coth t_n = n\tanh(nt_n). \tag{10}$$

The function ϕ_n increases on $(0, t_n)$, decreases on (t_n, ∞), and tends to zero at both ends. Define $b_{n,j} = 2\sin(j\pi/n)\phi_n(t_n)$, for $1 \leq j < n$.

Proposition 10. *For $n \geq 2$ and $\varepsilon > 0$, let $f_{n,\varepsilon}(z) = \varepsilon T_n(z) - 2\operatorname{Im} z$. Then*

$$N(f_{n,\varepsilon}) = n + 2\#\{1 \leq j < n : \varepsilon < b_{n,j}\} + \#\{1 \leq j < n : \varepsilon = b_{n,j}\}. \tag{11}$$

Every zero is nondegenerate unless $\varepsilon = b_{n,j}$ for some j. At such a parameter there is exactly one degenerate zero for each index j with $\varepsilon = b_{n,j}$.

Proof. On the real axis, $f_{n,\varepsilon}$ has the n simple zeros of T_n. At each such point x, its Jacobian is $\varepsilon^2 T_n'(x)^2 > 0$. On the upper branch $z_j(t)$, its real part vanishes precisely when $\varepsilon(-1)^j = 2\sin\theta_j\phi_n(t)$. On the reflected lower branch, the corresponding equation is $\varepsilon(-1)^j = -2\sin\theta_j\phi_n(t)$. Only the upper branch is possible for even j, and only the lower branch for odd j. On the selected branch, the equation reduces to

$$\varepsilon = 2\sin\theta_j\,\phi_n(t). \tag{12}$$

The monotonicity established before (10) gives two, one, or no solutions according as $\varepsilon < b_{n,j}$, $\varepsilon = b_{n,j}$, or $\varepsilon > b_{n,j}$. This proves (11).

The derivative of the imaginary part in the transverse θ-direction is, up to sign, $\varepsilon n(-1)^j \sinh(nt) \neq 0$. Along the selected branch, the derivative of the real part at a zero is a nonzero factor times $\phi_n'(t)$. In the regular coordinates (8), these derivatives prove nondegeneracy except at $t = t_n$. At t_n, the differential has rank one. Moreover, $\phi_n''(t_n) < 0$, because the logarithmic derivative is strictly decreasing. The two roots of (12) therefore meet in a quadratic tangency, and the stated degenerate zeros are the only ones. □

We now prove the explicit assertion of Corollary 5. The minimum of the $b_{n,j}$ is $b_{n,1}$, and

$$b_{n,1} \geq \frac{2\sin(\pi/n)\sinh(1/n)}{\cosh 1} \geq \frac{4}{n^2 \cosh 1} > \frac{1}{n^2}. \tag{13}$$

Here $\sin(\pi/n) \geq 2/n$, $\sinh(1/n) \geq 1/n$, and $\cosh 1 < 4$. Equations (11) and (13) show that $f_{n,1/n^2}$ has $n + 2(n-1)$ nondegenerate zeros. Finally, $-if_{n,1/n^2} = z - iT_n(z)/n^2 - \bar{z} = p_n(z) - \bar{z}$. Multiplication by $-i$ preserves zeros and nondegeneracy, completing the proof.

The location of the additional roots explains why the linearity of T_1 does not reduce the count. On every selected branch, one solution of (12) approaches $t = 0$, and the other approaches infinity as ε decreases to zero. Thus there are $n-1$ roots near the real critical points of T_n and $n-1$ roots escaping to infinity, in addition to its n real roots.

Figure 1 illustrates the count when $n = 5$.

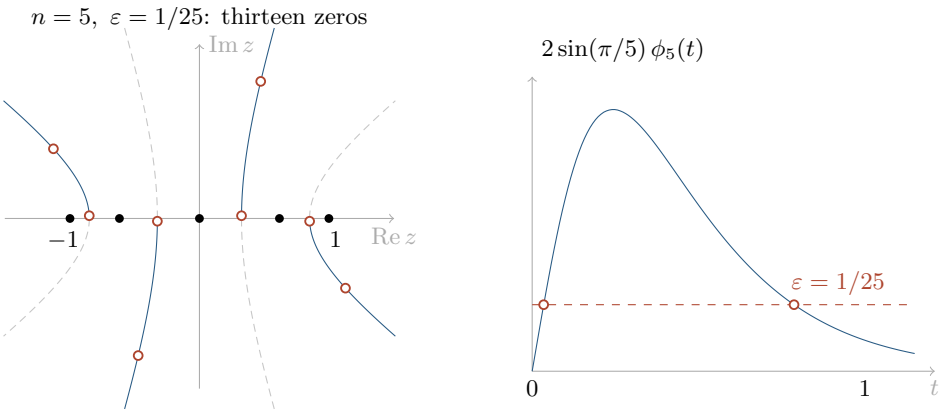


FIGURE 1. The degree-five example. Filled dots are the five real zeros; open circles are the eight nonreal zeros. Solid curves are the selected branches of $\operatorname{Im} T_5 = 0$; their conjugates are dashed. The right panel shows the two intersections that occur on each selected branch, here for $j = 1$ and $j = 4$.

2.3. **Intersections of translated branches.** We next prove Theorem 4. At height $y > 0$, the real coordinate of a branch with angle θ is $X_\theta(y) = \cot\theta\sqrt{y^2 + \sin^2\theta}$. The upper branches of $\operatorname{Im} T_m(z - a) = 0$ have real coordinates $a + X_{\varphi_\ell}(y)$, where $\varphi_\ell = \ell\pi/m$, $1 \leq \ell < m$.

Lemma 11. *Let $a > 2$ and $0 < \theta, \varphi < \pi$. The curves $x = X_\theta(y)$ and $x = a + X_\varphi(y)$, $y > 0$, have exactly one intersection if $\theta < \varphi$, and no intersection otherwise. Every intersection is transverse.*

Proof. For fixed y, $X_\theta(y)$ is strictly decreasing in θ: write it as $\cos\theta\sqrt{1 + y^2/\sin^2\theta}$ and differentiate. Thus $\theta \geq \varphi$ gives no intersection. If $\theta < \varphi$, the function $X_\theta(y) - X_\varphi(y)$ starts at $\cos\theta - \cos\varphi \in (0, 2)$ and tends to infinity. It is strictly increasing for $y > 0$. To see this, note that $X_\theta'(y) = y\cot\theta/\sqrt{y^2 + \sin^2\theta}$, and

$$\frac{\partial}{\partial\theta}\frac{\cot\theta}{\sqrt{y^2 + \sin^2\theta}} = -\frac{1}{\sin^2\theta\sqrt{y^2 + \sin^2\theta}} - \frac{\cos^2\theta}{(y^2 + \sin^2\theta)^{3/2}} < 0.$$

The equation $X_\theta(y) - X_\varphi(y) = a$ has exactly one solution, at which its derivative is positive. □

For the branch z_j in (9), let $k_j = \#\{1 \leq \ell < m : j/n < \ell/m\}$. Lemma 11 says that these are exactly the simple zeros of $\operatorname{Im} T_m(z_j(t) - a)$ for $t > 0$. There are $g - 1$ equality pairs $j/n = \ell/m$, where $g = \gcd(m, n)$; reflection $(j, \ell) \mapsto (n - j, m - \ell)$ interchanges the strict inequalities. Hence

$$\sum_{j=1}^{n-1} k_j = \frac{(n-1)(m-1) - (g-1)}{2}. \tag{14}$$

2.4. **Counting small levels.** Put $r(z) = T_m(z-a)$ and $A_j(t) = 2\operatorname{Im} r(z_j(t))/\cosh(nt)$. The upper branch requires $A_j(t) = \varepsilon(-1)^j$, and the lower branch requires $A_j(t) = -\varepsilon(-1)^j$. Together these are the two levels $A_j(t) = \pm\varepsilon$.

At the finite endpoint, $A_j(t) = 2r'(\cos\theta_j)\sin\theta_j\, t + O(t^2)$. Its linear coefficient is nonzero: the roots of T_m' lie in $(-1,1)$, whereas $\cos\theta_j - a < -1$; for $m = 1$ the derivative is one. At the other end, expand $r = \sum_{k=0}^m c_k T_k$, with real coefficients. Then $\operatorname{Im} r(z_j(t)) = \sum_{k=1}^m c_k \sin(k\theta_j)\sinh(kt)$. This sum is not identically zero, as its derivative at zero is nonzero. Let κ_j be the largest index for which $c_k \sin(k\theta_j) \neq 0$. Since $1 \le \kappa_j \le m < n$, there is a nonzero real constant C_j such that

$$\begin{aligned} A_j(t) &= C_j e^{-(n-\kappa_j)t}(1+o(1)), \\ A_j'(t) &= -(n-\kappa_j)C_j e^{-(n-\kappa_j)t}(1+o(1)). \end{aligned} \tag{15}$$

These estimates follow by differentiating the finite exponential sums, so the derivative estimate is justified separately. In particular, A_j is eventually strictly monotone and tends to zero. The index κ_j is needed when $\sin(m\theta_j) = 0$.

We now complete the proof of Theorem 4.

Proof. Step 1: counting each pair of branches. Choose disjoint small intervals around the k_j simple interior zeros of A_j, an initial interval on which $A_j' \neq 0$, and a terminal interval on which (15) applies. Each interior interval contains one solution to each level $A_j = \varepsilon$ and $A_j = -\varepsilon$, for all sufficiently small $\varepsilon > 0$. The initial and terminal intervals each contain exactly one solution to one of the two levels. The remaining set is compact and contains no zero of A_j; its positive minimum of $|A_j|$ excludes any further solution. Thus the pair contributes exactly $2k_j + 2$ simple solutions.

Step 2: completeness and regularity. On the real axis the zeros are exactly the n simple roots of T_n. Their Jacobians equal $|\varepsilon T_n'(x) + ir'(x)|^2 - |ir'(x)|^2 = \varepsilon^2 T_n'(x)^2 > 0$. At a nonreal zero, the real derivative along its branch is a nonzero factor times $A_j'(t)$, and the imaginary derivative transverse to the branch is, up to sign, $\varepsilon n(-1)^j \sinh(nt) \neq 0$. The coordinates (8) are regular, so the planar Jacobian is nonzero. A common sufficiently small parameter works for all $n-1$ pairs. The complete description of the real-level set after (9), followed by (14), gives

$$N(h_\varepsilon) = n + \sum_{j=1}^{n-1}(2k_j + 2) = (m+2)n - m - g.$$

This proves the theorem. □

The examples also give a uniform collection of counterexamples to the original Wilmshurst conjecture.

Corollary 12. *For $2 \le m \le n-2$, with $(m,n) \neq (2,4)$, there is a harmonic polynomial of degrees (n,m) with more than $3n - 2 + m(m-1)$ distinct zeros.*

Proof. Subtracting the proposed count from Theorem 4 gives $(m-1)(n-m-1)+1-\gcd(m,n)$. If $n-m \ge 3$, this is at least $(m-1)(n-m-2) > 0$. If $n-m = 2$, it equals $m - \gcd(m,2)$, which is positive except for $m = 2$. □

3. Perturbation and intersection bounds

For $f \in \mathbb{C}[z,w]$, its real Jacobian on $w = \bar z$ is

$$J_f(z) = |f_z(z,\bar z)|^2 - |f_w(z,\bar z)|^2. \tag{16}$$

A zero is *nondegenerate* if $J_f \neq 0$. The real differential has rank zero exactly when $f_z = f_w = 0$. Nondegenerate zeros persist uniquely under sufficiently small coefficient perturbations. A constant perturbation need not preserve all isolated zeros, so a different perturbation is needed.

We use the following standard form of parametric transversality. All manifolds in its statement are finite dimensional and second countable.

Lemma 13 ([Hir76, Chapter 3, Section 2])**.** *Let B, Y, Z be smooth manifolds, let $W \subset Z$ be a smooth submanifold, and let $\mathcal{F} : B \times Y \to Z$ be a smooth map transverse to W. For a residual set of $b \in B$, the map $\mathcal{F}_b : Y \to Z$ is transverse to W.*

The local observation below explains the use of opposite parameters.

Lemma 14. *Let $F : (\mathbb{R}^2, 0) \to (\mathbb{R}^2, 0)$ be real analytic with an isolated zero and* $\operatorname{rank} DF(0) = 1$. *Let G be real analytic near zero, with $G(0) \notin \operatorname{im} DF(0)$. For every sufficiently small $t > 0$, the total number of nondegenerate zeros of $F + tG$ and $F - tG$ in a fixed neighborhood of zero is at least two. The assertion is uniform when G varies in a sufficiently small finite-dimensional neighborhood.*

Proof. Real analytic source coordinates and an invertible real linear target change put F in the form $F(x, y) = (x, g(x, y))$, with $Dg(0) = 0$. Isolation implies $g(0, y) = cy^\ell + O(y^{\ell+1})$, where $c \neq 0$ and $\ell \geq 2$. Write $G = (G_1, G_2)$ in these coordinates and put $\beta = G_2(0) \neq 0$. The first equation of $F \pm tG = 0$ has a unique solution $x = X_\pm(y, t) = O(t)$. Substitute $y = t^{1/\ell}u$ into the second equation and divide by t. The resulting functions converge in C^1, on bounded u-intervals, to $cu^\ell \pm \beta$. Indeed, every mixed term of g contains x and has order at least two; after substitution its contribution, and its u-derivative, tend to zero. The same holds for the nonconstant terms of G_2. For odd ℓ, each limiting equation has one simple real root. For even ℓ, one has two simple real roots. The implicit function theorem preserves these roots and the nonvanishing two-dimensional Jacobian. The argument is uniform under the stated variation of G. □

Lemma 15. *Let $f \in \mathbb{C}[z, w]$ have finitely many zeros on $w = \bar{z}$. Every coefficient neighborhood of f contains a polynomial $\widetilde{f}$ such that $\deg_z \widetilde{f} \leq \deg_z f$, $\deg_w \widetilde{f} \leq \deg_w f$, and $N(\widetilde{f}) \geq N(f)$, with all zeros nondegenerate. The perturbation may be taken in any complex linear coefficient space containing f and stable under ∂_z, ∂_w.*

Proof. Step 1: generic perturbations away from rank-zero zeros. Set $S_f = \{z : f = f_z = f_w = 0 \text{ at } (z, \bar{z})\}$, which is finite, and let $f_\xi = f + \xi_1 f_z + \xi_2 f_w$, with $\xi = (\xi_1, \xi_2) \in \mathbb{C}^2$. At a zero of f_ξ outside S_f, at least one of f_z, f_w is nonzero. The differential with respect to ξ is therefore onto $\mathbb{C}$. Lemma 13 shows that, for a residual set of ξ, all zeros outside S_f are nondegenerate. We impose this condition on both ξ and $-\xi$. The zero sets are finite: if a real algebraic zero set with singularities confined to a finite set were infinite, its complement of that finite set would be an infinite discrete semialgebraic set, which is impossible.

Step 2: a choice of sign preserving the count. At every rank-one zero, the map $(a, b) \mapsto af_z + bf_w$ is onto $\mathbb{C}$. Choose $(a, b) \in \mathbb{C}^2$ so its value avoids the one-dimensional real image of dF_f at each such zero. Only finitely many proper real linear subspaces are excluded. Lemma 14, in disjoint neighborhoods of the rank-one zeros, gives at least two nondegenerate zeros in total for the two signs. Rank-two zeros persist for both signs, and the points of S_f remain zeros for both signs. Uniformity in Lemma 14 gives an open cone of small parameters with these properties. We may choose ξ in that cone satisfying Step 1; then

$$N(f_\xi) + N(f_{-\xi}) \geq 2N(f). \tag{17}$$

Step 3: termination. At each point of S_f, let H_k be the first nonzero homogeneous term of the complex Taylor expansion of f; here $k \geq 2$. Require also $a(H_k)_z + b(H_k)_w \not\equiv 0$. This excludes finitely many further proper complex linear subspaces. For every nonzero parameter in a suitable smaller cone, the vanishing order of $f_{\pm\xi}$ at each of these points is $k - 1$. Choose a sign in (17) with at least $N(f)$ zeros, and repeat. After the first step, every degenerate zero belongs to the original finite set S_f. At any point that remains rank zero the vanishing order strictly decreases. Once it reaches one, that point has rank one or two, and a further step removes any remaining degeneracy. The procedure therefore terminates. Successively smaller perturbations keep the final polynomial in the prescribed neighborhood. Derivatives increase neither degree, and every operation stays in the indicated coefficient space. □

The argument has reduced the upper-bound problem to generic polynomials. Let $X = \mathbb{CP}^1_z \times \mathbb{CP}^1_w$, with involution $\tau(z, w) = (\bar{w}, \bar{z})$ and fixed sphere L. An affine polynomial with $\deg_z f \leq n, \deg_w f \leq m$ is a section of $\mathcal{O}_X(n, m)$.

Lemma 16. *For $n, m \geq 1$, a dense set of sections of $\mathcal{O}_X(n, m)$ have a smooth connected zero curve C which avoids (∞, ∞) and is transverse both to L and to τC.*

Proof. The complete linear system separates first jets, and separates the values at any two distinct points. Its evaluation map is a submersion, so Lemma 13 gives smoothness and transversality to L. Off L, independent evaluation at $x, \tau x$ gives transversality of C and τC. At a fixed point, failure of this transversality would make the complex tangent line T_xC invariant under $d\tau_x$. An antilinear involution on a complex line has a nonzero fixed vector; this would contradict $T_xC \cap T_xL = \{0\}$. Avoiding (∞, ∞) excludes the hyperplane on which the coefficient of $z^n w^m$ vanishes.

A disconnected smooth divisor has disjoint irreducible components. If two components have bidegrees (n_i, m_i) and (n_j, m_j), their intersection number is $n_i m_j + m_i n_j$. All these integers are nonnegative. Pairwise disjointness would force all components into the same ruling, contrary to $n, m > 0$. Hence C is connected. □

We will also use the standard genus and intersection formulas.

Lemma 17 ([Har77, Chapter V, Proposition 1.5])**.** *A smooth connected curve of bidegree (n, m) in $\mathbb{CP}^1 \times \mathbb{CP}^1$ has genus $(n-1)(m-1)$. Two curves of bidegrees (n_1, m_1) and (n_2, m_2), without a common component, have intersection number $n_1 m_2 + m_1 n_2$.*

The intersection formula also underlies the coprime bound of Khavinson, Lundberg, and Perry [KLP24, Proposition 1.10]. Together, Lemmas 15–17 give $N(f) \le n^2 + m^2$ whenever $N(f) < \infty$: after regularization, a sufficiently small perturbation satisfying Lemma 16 preserves all displayed zeros, and its curves $C, \tau C$ meet in $n^2 + m^2$ points. This proves the quadratic term of (1) without a common-factor assumption.

For later constructions, the following total-degree version also proves finiteness and detects nondegeneracy at equality.

Lemma 18. *If $f(z, w) = cz^d + q(z, w)$, with $c \neq 0$ and $\deg_{\mathrm{tot}} q < d$, then $N(f) \le d^2$. If f has d^2 distinct zeros, all are nondegenerate.*

Proof. In a graded monomial order the leading monomials of f and f^* are z^d, w^d, after normalization. They are coprime, so these two polynomials form a Gröbner basis. The quotient has basis $\{z^i w^j : 0 \le i, j < d\}$ and length d^2. Every real-slice zero is an intersection of f, f^*. If there are d^2 distinct such points, each local intersection has length one. Its Jacobian determinant is the nonzero number in (16). □

4. The genus bound

The topological input is the Thom genus theorem.

Lemma 19 ([KM94, Theorem 1])**.** *Let Σ be a smooth closed connected oriented surface embedded in $\mathbb{CP}^2$, representing $d[\mathbb{CP}^1]$ with $d \ge 1$. Then $g(\Sigma) \ge (d-1)(d-2)/2$.*

By Lemmas 15 and 16, it suffices to consider a smooth connected curve $C \subset X$ of bidegree (n, m), avoiding (∞, ∞) and transverse to $L, \tau C$. Let $N = \#(C \cap L)$. The symmetric-product map

$$\Phi : X \longrightarrow \mathrm{Sym}^2(\mathbb{CP}^1) \simeq \mathbb{CP}^2, \qquad (z, w) \longmapsto \{z, \bar{w}\} \tag{18}$$

has fibers equal to the τ-orbits. In homogeneous coordinates it is $[z_0 : z_1], [w_0 : w_1] \mapsto [z_0\bar{w}_0 : z_1\bar{w}_0 + z_0\bar{w}_1 : z_1\bar{w}_1]$.

Lemma 20. *The restriction $\Phi|_C$ is an immersion with exactly $(n^2 + m^2 - N)/2$ transverse positive double points and no other multiple points.*

Proof. Away from L, the map is a local diffeomorphism. Near L, put $u = (z + \bar{w})/2$, $v = (z - \bar{w})/2$; in suitable target coordinates, $\Phi(u, v) = (u, v^2)$. If J is the complex structure of X, then $\ker d\Phi = J(TL)$ along L. The complex line TC is transverse to TL, hence also to $J(TL)$. Thus $\Phi|_C$ is immersive there.

Distinct points of C have the same image exactly when they are $x, \tau x$, with $x \in (C \cap \tau C) \setminus L$. The two branches are transverse, and no fiber has more than two points. Lemma 17 gives $\#(C \cap \tau C) = n^2 + m^2$; removing its N fixed points and pairing the rest proves the count. For the sign, transport the second branch to x by τ. This reverses its complex orientation, while Φ reverses the ambient four-dimensional orientation off L: it is the holomorphic symmetric-product map composed with conjugation in one factor. The two reversals cancel, leaving the positive intersection sign of two complex curves. □

We now prove the upper bound in Theorem 2.

Proof. The quadratic bound was proved after Lemma 17. For the linear bound, resolve every double point of Lemma 20 by the oriented local smoothing of two transverse complex coordinate disks. Each smoothing replaces two disks by an annulus, preserves the homology class, and adds one to the genus. This gives an embedded connected oriented surface $\Sigma \subset \mathbb{CP}^2$ with

$$g(\Sigma) = (n-1)(m-1) + \frac{n^2 + m^2 - N}{2}. \tag{19}$$

Let α, β be the positive degree-two cohomology generators of the two factors of X, and H the hyperplane class of $\mathbb{CP}^2$. Restriction of (18) to the factors gives $\Phi^* H = \alpha - \beta$. Since C has divisor class $n\alpha + m\beta$, its image has degree $m - n$. Reverse the orientation of Σ to obtain degree $d = n - m > 0$. Lemma 19 and (19) imply

$$(n-1)(m-1) + \frac{n^2 + m^2 - N}{2} \ge \frac{(n-m-1)(n-m-2)}{2}.$$

Rearranging gives $N \le (4m+1)n - 5m$. The initial perturbations did not decrease the zero count, so the same estimate holds for the original polynomial. □

Every value used to define $M_{m,n}$ is an integer bounded by (1), and the class is nonempty, for example $z^n w^m + 1$. Hence the maximum exists. Lemma 15, followed by Lemma 16, shows that it is attained with all zeros nondegenerate and a nonzero $z^n w^m$ coefficient. For such a polynomial, the mapping degree at infinity is $n - m$, so its positive and negative Jacobian counts satisfy $N_+ - N_- = n - m$. In particular, $M_{m,n} \equiv n - m \pmod 2$.

5. Radial constructions and sharpness

5.1. **Real roots from separated coefficients.** Our constructions use an elementary coefficient-dominance argument related to sign-independent real-rootedness. The latter was studied by Passare, Rojas, and Shapiro [PRS11] and by Forsgård, Novikov, and Shapiro [FNS17, Proposition 1]. We give the short form needed here, including gaps in the exponents.

For a list of nonzero real numbers, let V be its number of sign changes.

Lemma 21. *Let $0 \le e_0 < \cdots < e_s$ be integers and $c_0, \dots, c_s$ nonzero real numbers. If $0 < \rho_0 < \cdots < \rho_s$ satisfy*

$$|c_i|\rho_i^{e_i} > \sum_{j \ne i} |c_j|\rho_i^{e_j} \qquad (0 \le i \le s),$$

then $\sum_{i=0}^{s} c_i r^{e_i}$ has $V(c_0, \dots, c_s)$ positive roots and $V((-1)^{e_0}c_0, \dots, (-1)^{e_s}c_s)$ negative roots. All nonzero real roots are simple.

Proof. At ρ_i the polynomial has sign $\operatorname{sgn} c_i$. The intermediate value theorem gives a different positive root for each sign change. Descartes' rule bounds the positive roots, counted with multiplicity, by the same number. Equality follows, and every root is simple. Apply this argument to the polynomial at $-r$ for the negative roots. □

One choice of scales works for many polynomials at once. If $0 < \mu \le |b_i| \le B$, set $c_i = b_i t^{e_i^2}$ and $\rho_i = t^{-2e_i}$, where $0 < t < 1$. Then

$$t^{e_i^2} \sum_{j=0}^{s} c_j \rho_i^{e_j} = b_i + \sum_{j \ne i} b_j t^{(e_j - e_i)^2}. \tag{20}$$

The remainder in (20) has absolute value at most sBt; thus $sBt < \mu$ is sufficient.

Lemma 22. *Let $a, b \ge 0$ and $k > 2b$ be integers. Among maps*

$$F(z) = B(|z|^2) + z^k A(|z|^2), \qquad \deg A = a, \quad \deg B = b, \quad B(0) \ne 0,$$

with finite zero set, the maximal zero count is $k(2a + 2b + 1)$. It is attained with $A, B \in \mathbb{Q}[t]$, with all zeros nondegenerate and A, B having no common root.

Proof. Step 1: the upper bound. For a univariate polynomial A, write $A^{\#}(t) = \overline{A(\bar{t})}$. A positive common root of A, B would give a circle of zeros. Thus the positive roots of $\Psi(t) = t^k A(t) A^{\#}(t) - B(t) B^{\#}(t)$ are precisely the squared radii of zeros of F. Each gives k arguments. The support of Ψ lies in the disjoint blocks $\{0, \dots, 2b\}$ and $\{k, \dots, k + 2a\}$. It has at most $2a + 2b + 2$ nonzero monomials, so Descartes' rule gives at most $2a + 2b + 1$ positive roots.

Step 2: attainment. Choose rational $0 < \eta < 1/(4(a + b + 2))$, and set

$$B(t) = \sum_{j=0}^{b} (-1)^j \eta^{4j^2} t^j, \qquad A(t) = \sum_{j=0}^{a} (-1)^{b+1+j} \eta^{(k+2j)^2} t^j. \tag{21}$$

The exponents in $B(r^2) \pm r^k A(r^2)$ are $0, 2, \dots, 2b, k, k + 2, \dots, k + 2a$. Lemma 21 and (20) give $a + b + 1$ positive simple roots for the plus sign and $a + b$ for the minus sign. Applied separately to the two blocks, the same lemma places all positive roots of $B(r^2)$ between 1 and η^{-4b}, and all positive roots of $A(r^2)$ between η^{-2k} and $\eta^{-2(k+2a)}$. When a block is constant it has no roots. The positive roots exhaust the degrees of A, B, and the intervals are disjoint because $k > 2b$. Hence A, B are coprime, and the two signs have no common positive root. Their product is $-\Psi(r^2)$, giving exactly $2a + 2b + 1$ simple radii.

Step 3: planar regularity. At a zero $z = re^{i\theta}$, put $s = e^{ik\theta} \in \{1, -1\}$. The real radial derivative is $(B(r^2) + sr^k A(r^2))' \ne 0$. The imaginary angular derivative is $skr^k A(r^2) \ne 0$; the other two entries of this polar Jacobian vanish. Thus every zero is nondegenerate. □

Taking $a = b = m$ and $k = n - m > 2m$ in Lemma 22 gives bidegree (n, m) and

$$(4m+1)(n-m) \le M_{m,n} \le (4m+1)n - 5m, \qquad n > 3m. \tag{22}$$

Division of (22) by n proves the limit assertion of Theorem 2.

5.2. **Quadratic counts with few coefficients.** The same radial construction gives the elementary local models needed for the remaining degree ranges.

Lemma 23. *For every integer $d \ge 1$, there exists $Q_d(z, w) = z^d + \sum_{i+j<d} c_{ij} z^i w^j$ with coefficients in $\mathbb{Q}(i)$, conjugate degree at most $\lfloor d/2 \rfloor$, and exactly d^2 nonzero nondegenerate real-slice zeros.*

Proof. For $d = 1$, take $Q_1 = z - 1$. For odd $d = 2r + 1 \ge 3$, apply Lemma 22 with $a = 0, b = r, k = d$. The resulting $Az^d + B(|z|^2)$ has d^2 nondegenerate zeros, none at zero. Its leading homogeneous term is Az^d, and its conjugate degree is r.

For even $d = 2r$, take A, B from (21) with $a = 0, b = r - 1, k = d - 1$, and consider $P(z) = Az^d + \bar{z}B(|z|^2)$. At $z = \rho e^{i\theta} \neq 0$, division by $\bar{z}$ gives $A\rho^{d-1}e^{i(d+1)\theta} + B(\rho^2) = 0$. The proof of Lemma 22 gives $d - 1$ simple positive radii, each with $d+1$ arguments. The polar derivative test in that proof proves nondegeneracy at these points. The origin is also nondegenerate, since its linear term is $B(0)\bar{z}$. Thus P has $(d-1)(d+1) + 1 = d^2$ nondegenerate zeros. Its conjugate degree is r.

Divide either construction by A. In the even case, add a sufficiently small nonzero rational constant. All d^2 zeros persist and the origin ceases to be a zero; Lemma 18 excludes further zeros. The leading term and conjugate-degree bound are unchanged. □

We prove Theorem 9 using these models.

Proof. Before the final constant perturbation in the proof of Lemma 23, the construction for $d = 2r + 1$ has $r + 2 = \lceil d/2 \rceil + 1$ monomials, and that for $d = 2r$ has $r + 1 = \lceil d/2 \rceil + 1$ monomials. All their coefficients are real rational. The proof of that lemma already gives d^2 nondegenerate zeros.

For the quartic in (7), write $z = re^{i\theta} \neq 0$ and divide by $\bar{z}$. The equation is $r^3 e^{5i\theta} + r^2 - 1/8 = 0$. There is one positive radius with $e^{5i\theta} = 1$, namely $(\sqrt{5} - 1)/4$, and two with $e^{5i\theta} = -1$, namely $1/2$ and $(1 + \sqrt{5})/4$. Each gives five arguments, so the count, including zero, is sixteen. The origin has Jacobian $-1/64$; the three positive radii are simple, so the same polar derivative test proves regularity elsewhere.

It remains to exclude two monomials. More generally, let $F(z) = \alpha z^{a_1}\bar{z}^{b_1} + \beta z^{a_2}\bar{z}^{b_2}$ have total degree d, with $\alpha\beta \neq 0$ and distinct exponent pairs. Put $d_j = a_j + b_j$, $k_j = a_j - b_j$. A nonzero root satisfies

$$r^{d_1 - d_2} e^{i(k_1 - k_2)\theta} = -\beta/\alpha. \tag{23}$$

If $d_1 = d_2$, such a root generates an entire ray; if $k_1 = k_2$, it generates an entire circle. Finiteness excludes these cases. Otherwise (23) gives one radius and $|k_1 - k_2| \le d_1 + d_2 \le 2d - 1$ arguments. Including the origin gives at most $2d$ zeros. A single monomial has at most one zero. For $d = 4$, both possibilities fall short of sixteen. □

The same argument shows that degree four is the first failure of Conjecture 8. Degrees one and two have the extremizers z and $z^2 - \bar{z}$. In degree three, the trinomial $z^3 + z\bar{z} - 1/8$ has nine zeros: its imaginary part vanishes on three lines, and its restriction, with a suitable orientation of each line, is $r^3 + r^2 - 1/8 = (r + 1/2)(r^2 + r/2 - 1/4)$. This has three distinct nonzero real roots, whereas a binomial has at most six planar zeros. Thus the minimum support sizes in degrees $1, 2, 3, 4$ are respectively $1, 2, 3, 3$.

5.3. **Splitting a zero and transferring zeros from infinity.**

Lemma 24. *Suppose $P(z, \bar{z})$ has a finite zero set, a unique monomial of highest total degree, and only nondegenerate zeros away from the origin. Suppose also that $P(z, \bar{z}) = c\bar{z}^s + O(|z|^{s+1})$ at zero, where $c \neq 0$ and $s \ge 1$. There are arbitrarily small perturbations by terms of total degree less than s that replace zero by exactly s^2 nondegenerate zeros and preserve exactly one zero near each other zero. If P has coefficients in $\mathbb{Q}(i)$, the perturbation may have coefficients in $\mathbb{Q}(i)$.*

Proof. Conjugate the polynomial Q_s constructed in Lemma 23 to obtain $H(z, \bar{z}) = \bar{z}^s + \sum_{i+j<s} h_{ij} z^i \bar{z}^j$ with s^2 nonzero nondegenerate zeros. For $t > 0$, define $P_t = P + c\sum_{i+j<s} h_{ij} t^{s-i-j} z^i \bar{z}^j$. After $z = t\zeta$ and division by ct^s, the maps converge in C^1 on compact sets to H. Every zero of H therefore gives one simple zero of P_t.

To exclude a further scale, choose a fixed small disk on which $|P(z)| \ge |c||z|^s/2$. For $|z| \ge Ct$ in that disk, the perturbation is bounded by $|c||z|^s \sum_{i+j<s} |h_{ij}| C^{-(s-i-j)}$. Choose C large enough that this sum is less than $1/2$, and that $|\zeta| < C$ contains the zeros of H. All new zeros in the disk lie in $|z| < Ct$. Compactness and local uniqueness in that scaled disk give exactly s^2 zeros. The other old zeros persist uniquely. On the

remaining compact set the original map has a positive minimum modulus, while the unchanged unique highest monomial gives a uniform nonvanishing estimate near infinity. This excludes all further zeros. Rational t retains the asserted coefficients. □

Proposition 25. *If $n > m \geq 1$ and a polynomial $f = z^n + q(z, w)$, with $\deg_{\mathrm{tot}} q < n$ and $\deg_w f \leq m$, has n^2 distinct zeros, then $M_{m,n} = n^2 + m^2$. If the given coefficients belong to $\mathbb{Q}(i)$, an extremizer with coefficients in $\mathbb{Q}(i)$ and all zeros nondegenerate exists.*

Proof. The zeros of f are nondegenerate by Lemma 18. Conjugate the polynomial Q_m of Lemma 23, writing $H(u, v) = v^m + \sum_{i+j<m} h_{ij} u^i v^j$. It has m^2 nonzero nondegenerate zeros and $h_{00} \neq 0$. Define

$$g_t(z, w) = f(z, w) + \sum_{i+j<m} h_{ij} t^{m-i-j} z^{n-i} w^{m-j}. \tag{24}$$

All exponents in (24) are nonnegative, and the coefficient of $z^n w^m$ is $h_{00} t^m \neq 0$. The n^2 old zeros persist as $t \to 0$. Under inversion, put $G_t(u, v) = u^n v^m g_t(1/u, 1/v)$. Every nonleading monomial of f contributes a term of total degree at least $m + 1$ to G_t. Consequently $t^{-m} G_t(t\zeta, t\eta) \to H(\zeta, \eta)$ in C^1 on compact real-slice sets. Its m^2 nonzero simple zeros yield m^2 zeros of g_t escaping to infinity. Inversion and multiplication by a nonvanishing factor preserve nondegeneracy.

For each sufficiently small t, a further arbitrarily small generic coefficient perturbation preserves these $n^2 + m^2$ zeros. Lemma 16 makes its full zero set finite, and (1) excludes any additional zeros. The nonzero corner coefficient makes both degrees exact. When the input is rational complex, take t rational and choose the final perturbation rational complex within the open set where these zeros persist and the complex intersection is transverse. □

We can now prove Theorem 3.

Proof. Step 1: the quadratic range. If $m < n \leq 2m+1$, then $\lfloor n/2 \rfloor \leq m$. The polynomial Q_n from Lemma 23 satisfies Proposition 25, which gives $M_{m,n} = n^2 + m^2$.

Step 2: a construction for $n \geq 2m + 1$. Put $k = n - m$ and $s = \max\{0, 3m + 1 - n\}$. Then $0 \leq s \leq m < k$ and $k - s > 2(m - s)$. Choose A, B by (21) with degrees $m, m - s$ and exponent $k - s$, and set $P(z) = z^k A(|z|^2) + \bar{z}^s B(|z|^2)$. For $z = re^{i\theta} \neq 0$, division by $\bar{z}^s$ gives $B(r^2) + r^{k-s} e^{i(k+s)\theta} A(r^2) = 0$. The proof of Lemma 22 gives $4m - 2s + 1$ simple positive radii and $k + s$ arguments at each radius. Its derivative test proves nondegeneracy.

The unique highest monomial is a nonzero multiple of $z^n \bar{z}^m$. If $s > 0$, the initial term at zero is $B(0)\bar{z}^s$, and Lemma 24 replaces it by s^2 simple zeros without altering the bidegree. If $s = 0$, the origin is not a zero. The complete count is

$$(k + s)(4m - 2s + 1) + s^2 = (4m + 1)(n - m) + (6m + 1 - 2n)s - s^2.$$

For $s > 0$, the last two terms are $s(s - 1)$; for $s = 0$, they vanish. This is (3).

Step 3: the uniform comparison. For $n \geq 2m+1$, (3) is at least $(4m+1)(n-m)$. Since $U_{m,n} \leq (4m+1)n - 5m$, their difference is at most $4m(m - 1)$, proving (2). The remaining range follows from Step 1. □

6. An exact maximum for a restricted harmonic family

We prove Theorem 7. The upper estimate refines the Descartes argument of Lundberg [Lun23, Section 4.1]. The lower estimate uses Lemma 21 simultaneously on every line; the coefficients of these restrictions are not independent.

Lemma 26. *Let $N > M \geq 0$, $a \in \mathbb{R} \setminus \{0\}$, and $u \in \mathbb{R}[r]$ have degree at most M. Put $b = [r^M]u$, where $[r^M]u$ denotes the coefficient of r^M. The number of nonzero real roots of $ar^N + u(r)$, counted with multiplicity, is at most $M + 1$ if $N - M$ is odd, and at most $M + 2\mathbf{1}_{\{ab<0\}}$ if $N - M$ is even. Here $\mathbf{1}_E$ is one when E holds and zero otherwise.*

Proof. The case $u = 0$ is immediate. Otherwise, let $d = \deg u$. For the coefficient lists of $u(r)$ and $u(-r)$, an odd gap between successive nonzero exponents contributes exactly one sign change in total. An even gap contributes zero or two, hence at most its size. The internal contribution is therefore at most d. Appending degree N contributes one if $N - d$ is odd, and two or zero according to the relative signs if $N - d$ is even. For $d = M$, Descartes' rule proves the result. If $d < M$ and $N - M$ is odd, use $d + 2 \leq M + 1$. If $N - M$ is even, $d = M - 1$ gives $d + 1 = M$, while $d \leq M - 2$ gives $d + 2 \leq M$. These cases also cover $b = 0$. □

6.1. **The upper bound.** A rotation of the variable makes $\varepsilon > 0$. Then the imaginary part of $h(z) = \varepsilon z^n + q(z) + \overline{q(z)}$ vanishes on the n distinct lines $\Lambda_j = e^{ij\pi/n}\mathbb{R}$, $0 \le j < n$. Write $q(z) = \sum_{k=0}^{m} a_k z^k$, with $a_m \neq 0$, and $\theta_j = j\pi/n$. On Λ_j, the restriction is the real polynomial

$$f_j(r) = \varepsilon(-1)^j r^n + 2\sum_{k=0}^{m} \operatorname{Re}(a_k e^{ik\theta_j}) r^k. \tag{25}$$

Its leading coefficient is nonzero, so every member of the family has finitely many zeros.

First assume $\operatorname{Re} a_0 \neq 0$. The origin is not a zero, and the line counts add without overlap. If $n - m$ is odd, Lemma 26 gives at most $m + 1$ roots per line; here $\delta_{m,n} = 0$. Suppose $n - m = 2d$. Put $G = \gcd(n, d)$ and $\ell = n/G$, and write $a_m = |a_m| e^{i\beta}$. The sign governing the final exponent gap is

$$(-1)^j \operatorname{Re}(a_m e^{imj\pi/n}) = |a_m| \cos\left(\beta - \frac{2\pi dj}{n}\right).$$

If B_β is the number of negative cosines in this list, Lemma 26 yields

$$N(h) \le nm + 2B_\beta, \qquad 2B_\beta \le \begin{cases} n, & \ell \text{ even}, \\ n + G, & \ell \text{ odd}. \end{cases} \tag{26}$$

Indeed, the arguments run through a regular ℓ-gon, each vertex repeated G times. An open semicircle contains at most $\ell/2$ vertices when ℓ is even, by antipodal pairing, and at most $(\ell + 1)/2$ when ℓ is odd, by their spacing. Vertices on the boundary are not counted.

To express (26) in terms of $\delta_{m,n}$, write $m = gu, n = gv$, where $g = \gcd(m, n)$ and $\gcd(u, v) = 1$. If u, v are odd, then $G = g \gcd(v, (v - u)/2) = g$, and $\ell = v$ is odd. Otherwise u, v have opposite parity. Since $n - m$ is even, g is even, and $G = (g/2)\gcd(2v, v - u) = g/2$, so $\ell = 2v$ is even. This proves the upper bound $n(m + 1) + \delta_{m,n}$.

If $\operatorname{Re} a_0 = 0$, divide every f_j by r and apply Lemma 26 with $(N, M) = (n - 1, m - 1)$. The last-gap parity and the relevant leading signs are unchanged, and each nonzero-root bound decreases by one. Adding the common origin only once gives the stronger bound $n(m + 1) + \delta_{m,n} - n + 1$. Thus the upper bound includes missing coefficients, singular zeros, and a zero at the origin.

6.2. **Rational coefficients attaining the bound.** Put $g = \gcd(m, n)$ and define $\eta = 1/(4n)$, $t = 1/(16n(m + 1))$, and $\gamma = 1 + i\eta$. Put $\sigma = -1$ when $\delta_{m,n} = g$ and $n/g \equiv 1 \pmod 4$; put $\sigma = 1$ otherwise. Our explicit choice is

$$q(z) = \frac{\sigma\gamma}{2} \sum_{k=0}^{m} t^{k^2} z^k, \qquad \varepsilon = t^{n^2}. \tag{27}$$

The coefficients in (27) are rational complex, the degrees are exact, and $h(0) = \sigma \neq 0$. The restriction (25) becomes

$$f_j(r) = (-1)^j t^{n^2} r^n + \sigma \sum_{k=0}^{m} \left(\cos\frac{kj\pi}{n} - \eta \sin\frac{kj\pi}{n}\right) t^{k^2} r^k.$$

For every k, j in these ranges,

$$\frac{1}{4n} \le \left|\cos\frac{kj\pi}{n} - \eta \sin\frac{kj\pi}{n}\right| < 2. \tag{28}$$

A zero cosine gives absolute value exactly η. A nonzero cosine has magnitude at least $\sin(\pi/(2n)) \ge 1/n$, because its argument is an integer multiple of π/n. Subtracting a term of magnitude at most η proves (28).

Apply (20) with support $\{0, 1, \ldots, m, n\}$, $\mu = 1/(4n)$, and $B = 2$. The remainder is at most $2(m + 1)t = 1/(8n) < \mu$. Lemma 21 therefore gives the exact Descartes count, with every root simple, on all the lines at once. The consecutive low exponents contribute exactly m roots on each line. When $n - m$ is odd, the last gap adds one more.

For even $n - m = 2d$, use the integers G, ℓ defined in the preceding subsection. Write $\gamma = |\gamma| e^{i\alpha}$, where $0 < \alpha = \arctan\eta < 1/(4n)$. The last gap adds two roots exactly when $\sigma\cos(\alpha - 2\pi dj/n) < 0$. If ℓ is even, antipodal pairing gives exactly $n/2$ negative cosines, none zero by (28). If ℓ is odd, then $G = g$ and $\ell = n/g$. At phase zero, a regular ℓ-gon has $(\ell - 1)/2$ vertices in the negative semicircle when $\ell \equiv 1 \pmod 4$, and $(\ell + 1)/2$ when $\ell \equiv 3 \pmod 4$. This is the count of integers strictly between $\ell/4$ and $3\ell/4$. Every vertex is at least $\pi/(2\ell)$ from the boundary, so the small phase α does not change these counts. The definition of σ selects $(\ell + 1)/2$ negative cosines, each repeated g times. The final gaps contribute $n + g$ roots in this case, and n in the even case. Adding the nm roots from the low exponents proves (6).

Finally, at a root $z = e^{i\theta_j} r_0$, use local coordinates $z = e^{i\theta_j}(r + is)$. The derivatives satisfy $\partial_r \operatorname{Re} h = f_j'(r_0) \neq 0$, $\partial_r \operatorname{Im} h = 0$, and $\partial_s \operatorname{Im} h = t^{n^2} n(-1)^j r_0^{n-1} \neq 0$. Thus all planar zeros are nondegenerate, completing the proof of Theorem 7.

For comparison, the Chebyshev count exceeds (6) by $n - m - g - \delta_{m,n}$, a nonnegative integer. If $\delta_{m,n} = 0$, use $g \mid (n - m)$. If $\delta_{m,n} = g$, the integer $(n - m)/g$ is positive and even. For fixed m, the improvement has size $n - O_m(1)$.

7. Further questions

The error in Theorem 3 is bounded independently of n. Determining $M_{m,n}$ in all degrees, including the full region where $M_{m,n} = n^2 + m^2$, remains open.

For each fixed $m \geq 2$, does $H_{m,n}/n$ have a limit? Equation (5) places any such limit between $m + 2$ and $4m + 1$. The orientation-reversing-zero conjecture of Lee, Lerario, and Lundberg [LLL15] would lower the upper coefficient to $2m + 1$.

Let s_d be the least number of nonzero monomials in a degree-d polynomial with d^2 distinct zeros. Theorem 9 gives $3 \leq s_d \leq \lceil d/2 \rceil + 1$ for $d \geq 3$, with $s_3 = s_4 = 3$. Determining s_d for $d \geq 5$ remains open.

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